Exercise 2: Line Detection using Hough Transform

Aim:

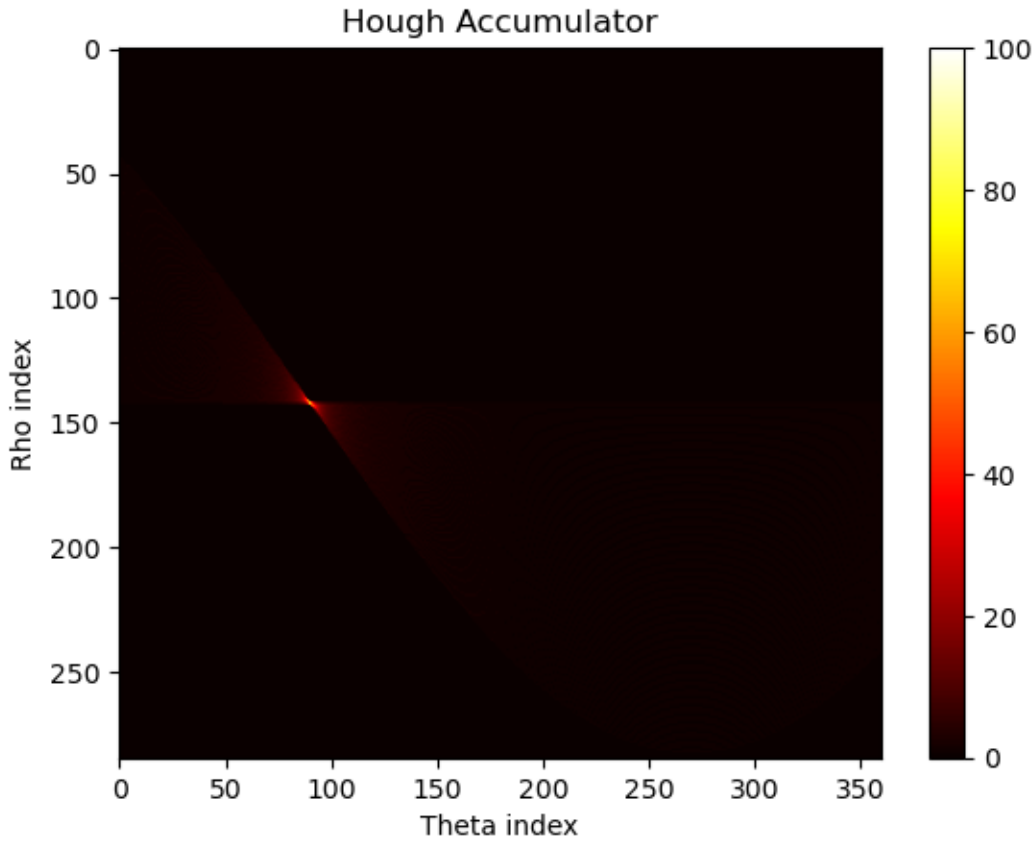
Detect straight lines in a binary edge image using the Hough Transform voting procedure in (ρ, θ) parameter space.

Code:

```
import numpy as np
from scipy import ndimage
import matplotlib.pyplot as plt

def hough_line_detection(binary_edges, theta_step=0.5, threshold_ratio=0.6):
    h, w = binary_edges.shape
    diag = int(np.ceil(np.hypot(h, w)))

    rhos = np.linspace(-diag, diag, 2 * diag + 1)
    thetas = np.deg2rad(np.arange(-90, 90, theta_step))

    accumulator = np.zeros((len(rhos), len(thetas)), dtype=int)

    y, x = np.nonzero(binary_edges > 0.1)

    for px, py in zip(x, y):
        for t_idx, theta in enumerate(thetas):
            rho = px * np.cos(theta) + py * np.sin(theta)
            rho_idx = np.argmin(np.abs(rhos - rho))
            accumulator[rho_idx, t_idx] += 1

    max_votes = accumulator.max()
    peaks = np.argwhere(accumulator >= threshold_ratio * max_votes)

    lines = []
    for rho_idx, theta_idx in peaks:
```

```
        lines.append((rhos[rho_idx], thetas[theta_idx]))

    return lines, accumulator



img = np.zeros((100, 100))
for i in range(100):
    img[i, i] = 1.0

lines, acc = hough_line_detection(img)

print("Detected lines (rho, theta rad):", lines)

plt.imshow(acc, cmap='hot', aspect='auto')
plt.xlabel("Theta index")
plt.ylabel("Rho index")
plt.title("Hough Accumulator")
plt.colorbar()
plt.show()
```

Output:


Hough Accumulator
Rho index
Theta index
0
50
100
150
200
250
0
50
100
150
200
250
300
350
100
80
60
40
20
0